The Developmental and Transcriptomic Identities of GLP-1R Cells

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Abstract

GLP-1 receptor agonists are used to treat metabolic disorders, yet the developmental origins of the cells they target remain poorly understood. Using publicly available data, we investigated GLP-1R-relevant cells across development and in the adult brainstem. We profiled incretin axis genes in a prenatal mouse single-cell atlas and compared Glp1r+ and Gfral+ transcriptomes in the dorsal vagal complex. Our analysis revealed distinct developmental trajectories, potency states, and transcriptomic identities.

Introduction

Glucagon-Like Peptide 1 (GLP-1) is secreted primarily from the gut and brain to regulate energy homeostasis, appetite, and insulin levels[1,2]. Given its strong effects on appetite, insulin, and weight regulation, it became an attractive therapeutic target to treat metabolic disease[1]. GLP-1 binds to GLP-1 receptors (Glp1r in mice) in the brain and gastrointestinal tract, and GLP-1 receptor agonists were developed to mimic these effects for therapeutic purposes. The use of pharmaceutical GLP-1 receptor agonists has grown in popularity, with an estimated over 9 million users in the United States[3]. These medications, such as Ozempic, Wegovy, and Mounjaro, are popularized for rapid weight loss results and treatment of Type II diabetes. As such, they are of wide interest to consumers, pharmaceutical companies, and researchers due to efficacy and demand. Between 2018 and 2023, spending on Ozempic increased from $410 million to $26.42 billion dollars[4]. Given the lucrative financial benefits and improved health outcomes of patients, it is important to understand the cells responsible for these powerful effects. In this study, we seek to understand the developmental trajectories of GLP-1R cells, which could unveil insights for developing an in vitro system to study their agonists.

To contextualize Glp1r+ cells across development, we use a large scale single-cell RNA sequencing atlas of mouse prenatal development that profiles over 11 million nuclei across embryos from gastrulation to birth, capturing hundreds of cell types across the entire organism[5]. While GLP-1R has received the most therapeutic attention, next-generation metabolic drugs are increasingly targeting multiple receptors in the incretin axis[6]. Tirzepatide is a dual GLP-1R/GIPR agonist[7], and triple agonists targeting GLP-1R, GIPR, and GCGR are in Phase 3 trials[8]. GLP-1 and glucagon are both encoded by Gcg, but are processed differently by tissue: intestinal L-cells produce GLP-1, whereas pancreatic alpha cells produce glucagon[9]. These hormones bind distinct receptors, GLP-1R and GCGR, while GIP binds to GIPR. As these therapies target multiple nodes in the same axis, studying GLP-1R alone is insufficient.

Using this prenatal atlas, we profiled Glp1r, Gcgr, Gipr, and Gcg across development to map their tissue distribution, temporal dynamics, and co-expression patterns. We then applied CytoTRACE2[10] to curated GLP-1R-relevant cell types to measure developmental potency and used PAGA[11] on a neuroendocrine subset to examine transcriptomic relationships among brain cell types relevant to GLP-1 signaling.

Additionally, many users of these drugs remain ill-informed about their mechanism of action, including how they function in the brain. In the brain, GLP-1R+ neurons are primarily found in the Dorsal Vagal Complex (DVC) of the medulla, a region of the brainstem largely responsible for regulating appetite[12].

More specifically, it has been shown that GLP-1 receptor neurons in the Area Postrema (AP) and the Nucleus of the Solitary Tract (NTS) are responsible for behavioral responses to GLP-1 agonists in mice[12]. These brainstem regions also contain Gfral receptor neurons, which respond to the cytokine GDF15. GDF15 is a biomarker of stress, including cancer, tissue injury, and pregnancy, but it is also known for its effects on appetite suppression, and high levels of GDF15 are linked to cachexia in disease[13]. Here, we use single nucleus RNA sequencing of the mouse DVC[14] to analyze Glp1r and Gfral neurons to explore transcriptomic differences between these cell populations to investigate their different roles in regulating appetite.

Results

GLP-1R Cell-Type Signature Analysis

To identify cell type specific transcriptional programs associated with Glp1r expression in the downsampled pilot dataset, we first ranked eligible cell types by the fraction of Glp1r+ cells. The three enriched populations were glomerular endothelial cells occurring at a rate of 13.9%, lung cells (Eln+) occurring at a rate of 8% and pancreatic islets occurring at a rate of 4.3% (Fix XX A). We then trained within cell type classifiers to distinguish Glp1r+ from Glp1r- using both L1 regularized logistic regression and XGBoost. Both methods achieved above baseline discrimination across all three populations. Here the strongest performance was in pancreatic islets: L1 AUROC 0.836, AUPRC 0.660; XGBoost AUROC 0.813, AUPRC 0.608. Following this we see Eln+ lung cells with L1 AUROC 0.729, AUPRC 0.566; XGBoost AUROC 0.732, AUPRC 0.556, and lastly with the lowest performance amongst the 3 cell types, we see glomerular endothelial cells with L1 AUROC 0.622, AUPRC 0.434; XGBoost AUROC 0.665, AUPRC 0.470.

Stable feature selection further indicated that Glp1r- associated signatures were highly cell type specific. The L1 model identified 251 stable non-zero genes whereas XGBoost identified 52 stable genes. Nearly all stable features were tissue specific. The L1 analysis produced 242 tissue specific genes and only 9 shared mixed genes, with no shared core genes. On the other hand all stable XGBoost features were tissue specific. Notably, a subset of genes were recovered by both features selection frameworks with concordant directionality. These include Ldb2 in glomerular endothelial cells; Adgrl3, Fam49b, Galnt18, Kit, Ptprm, and Scn7a in lung cells (Eln+); and Alcam and Ins2 in pancreatic islets. These cross model overlaps highlight a small group of candidate genes that appear robust to modelling approaches.

Developmental trajectories of incretin axis genes

To determine when and where incretin axis genes are expressed during prenatal development, we analyzed pseudobulk expression profiles across all cell types in the Qiu et al. atlas. The four genes showed distinct tissue distributions. Glp1r expression was the highest in enteric neurons and glomerular endothelial cells, consistent with known adult expression in the gut nervous system and kidney vasculature. Gcgr expression was the highest in retinal ganglion cells rather than hepatocytes, which is the primary adult target of glucagon signaling. Gipr showed the broadest expression, with ventricular cardiomyocytes as the top-expressing cell type at mean levels almost 3 times higher than any other cell type in all the panels. Gcg was concentrated in pancreatic islets and intestinal enteroendocrine cells, as expected.

Recruitment across developmental time further separated these genes (Figure 1A). Gipr was expressed in 12 cell types at E8.5 and increased to over 55 by mid-gestation, consistent with broad developmental recruitment. Glp1r remained restricted to 2-8 cell types across the prenatal period, indicating a highly selective and temporally restricted expression. A heatmap of Glp1r expression across time (Figure 1B)

shows that these few cell types expressed Glp1r in narrow timepoint windows, rather than continuously. Extraembryonic visceral endoderm was a notable outlier, with very high Glp1r expression around E11.5, and was excluded because it is not part of the embryo proper.

CytoTRACE2 showed a biologically coherent developmental potency hierarchy across 12 curated GLP-1R-relevant cell types (Figure 1D). Primordial germ cells scored the highest at approximately 0.85, serving as a positive control, whereas intestinal enteroendocrine cells, which produce GLP-1, scored near zero. In contrast, hypothalamus and pituitary gland cells, both targets of GLP-1 signaling in the brain, retained intermediate potency. Olfactory epithelial cells ranked second overall, consistent with neurogenesis in the olfactory epithelium. Only 10 of 720 sampled cells were detected as Glp1r+, highlighting the dropout of this low-expression gene and supporting the use of pseudobulk profiling.

PAGA analysis of a neuroendocrine subset identified the hypothalamus as the central hub among GLP-1-responsive populations (Figure 1C). Parasympathetic neuron and cochlear auditory hair cell clusters remained distinct, while hypothalamus, pituitary, olfactory epithelial, and enteric neuron populations formed a connected manifold. The strong graph connectivity linked the hypothalamus to the olfactory epithelial and pituitary gland cells.

Glp1r+ and Gfral+ cells in the Dorsal Vagal Complex

Because Glp1r and Gfral are related to appetite suppression in different contexts, we sought to investigate the genetic identities of cells expressing these receptors. We used an existing single nucleus RNA sequencing dataset composed of cells isolated from the dorsal vagal complex of 30 mice. After completing quality control by ensuring that cells with high mitochondrial gene counts (indicating poor cell viability) were removed, we subsetted this dataset to all cells expressing Glp1r and Gfral and performed clustering by cell type to identify the different cell types that express these receptors (Figure 2A). We identified 1358 Glp1r receptor expressing cells (Glp1r+), 694 Gfral receptor expressing cells (Gfral+), and 52 cells that express both. As with the original DVC dataset[14] expression of both Glp1r and Gfral were consistently high in monoamine neurons which synthesize norepinephrine and serotonin (Figure 2B). This corresponds to the M0 neuronal class based on the original dataset annotations, which are biologically responsible for also producing neuromodulators like serotonin and norepinephrine. Because serotonin is involved in mood and behavioral regulation, the enrichment of these specific M0 neurons suggests that the GLP-1 receptor pathways potentially influence processes beyond appetite. Interestingly, Glp1r expression was also high in *Sall3* neurons, while Gfral expression was not (Figure 2B).

Differential gene expression between Glp1r+ and Gfral+ cells

We then performed a differential expression analysis between Glp1r+ and Gfral+ cells across all cell types in our initial DVC subset and found very few significant differentially expressed genes (Figure 3A). However, among the 9 differentially expressed genes, *SLC6A2*, which was upregulated in Glp1r+ cells, was of particular interest.

Because neurons in the dorsal vagal complex (DVC) are largely responsible for the appetite suppression effects of GLP-1 and GDF15[12], we next investigated the genetic identities of only neurons. We performed a differential expression analysis to compare Glp1r+ and Gfral+ neurons and identified 10 differentially expressed genes (Figure 4A). Many were the same as the initial analysis with the full cell population, but we did identify *Lcp2* slightly upregulated in Gfral+ neurons.

Across both levels of granularity, only one Hallmark gene set remained significant after Benjamini-Hochberg false discovery rate correction[15]. Namely, the *Hallmark KRAS signalling up* gene set

was significantly enriched in Glp1r+ cells when compared to Gfral+ cells with an adjusted p-value of 0.006 and a normalized enrichment score of 1.97, the highest NES among all gene sets tested across all levels (Figure 3B, 3C).

Although not statistically significant after p-value correction, gene sets that were higher in Glp1r+ neurons relative to Gfral+ neurons with the highest magnitude normalized enrichment score were *Hallmark inflammatory response, Hallmark angiogenesis, and Hallmark KRAS signalling down* (Figure 4B). While gene sets of the highest magnitude in Gfral+ neurons relative to Glp1r+ neurons were once again *Hallmark KRAS signalling up* and *Hallmark hedgehog signalling* (Figure 4B).

To better understand the genes co-expressed with both Glp1r+ and Gfral+ cells, Spearman correlation was conducted with each expressed gene to either receptor with a correlation bias score calculated as *corr(gene, Glp1r) - corr(gene, Gfral).* Thus, a positive score corresponds with an increase in Glp1r expression and a negative score corresponds with an increase in Gfral expression. Functional GO analysis of the top 100 genes yielded enrichment for neuronal regulation pathways in Gfral and glutamate receptor signaling pathways in Glp1r (Figure 3D).

Discussion

These findings from the downsampled pilot dataset discussed in the "GLP-1R Cell-Type Signature Analysis" section suggest that transcriptional correlations of Glp1r expressions are highly context-dependent rather than defined by a single shared program. The predominance of tissue specific features, together with minimal overlap across the three top-ranked cell types suggests that Glp1r positivity is associated with distinct molecular states in different cellular contexts. The similar performance of XGboost and L1 regularized models further suggests that much of this signal can be captured by relatively compact feature sets. Genes recovered by both approaches including: Adgrl3, Alcam, Kit, and Ins2 therefore represent a subset of strong candidates for downstream study. Interestingly Adgrl3, Alcam and Kit are involved in forming neuronal adhesion, synapse organization, and in general signaling pathways that influence neuronal connectivity [16]. These functions are particularly relevant given that GLP-1 signaling operates within the gut brain circuits that link peripheral metabolic signals to brainstem neurons in the dorsal vagal complex. Ins2, one of the genes encoding insulin in mice, further highlights the metabolic context of GLP-1 signalling. GLP-1 is well known to enhance glucose dependent insulin secretion from pancreatic β cells and its receptor is expressed in tissues involved in coordinating systemic metabolic responses[17]. Thus the presence of ins2 among the predictive features in pancreatic islets likely reflects the close physiological coupling between GLP-1 signaling and insulin regulation.

Our profiling of the full incretin axis across prenatal development produced three main findings. First, Gipr, not Glp1r, was the dominant developmental receptor in this axis, appearing in over 55 cell types by mid-gestation, including ventricular cardiomyocytes (Figure 1A). This may be relevant to the safety profile of dual and triple agonists such as tirzepatide and retatrutide, which target GIPR as well as GLP-1R. The developmental role of GIPR in cardiac tissue remains unclear.

Second, Glp1r expression was developmentally restricted and temporally specific, turning on in a small number of cell types at defined gestational windows (Figure 1B). The earliest strong signals appeared in olfactory pit cells and primordial germ cells, while later expression emerged in enteric neurons and glomerular endothelial cells. These patterns suggest that several known adult GLP-1R-expression populations are established prenatally.

Third, CytoTRACE2 revealed a potency asymmetry: cells that produce GLP-1, such as intestinal enteroendocrine cells, are terminally differentiated, while cells that respond to GLP-1 in the brain, such as hypothalamus and pituitary cells, retain intermediate developmental potency (Figure 1D). Because less

differentiated cells may be more responsive to signaling cues, this difference raises the possibility that developmental exposure to GLP-1 agonists could affect tissues differently depending on their maturation state.

This difference suggests that GLP-1-producing and GLP-1-responsive tissues occupy distinct developmental states, which may influence how these systems respond to metabolic signaling during development. Consistent with this, PAGA placed the hypothalamus at the center of the neuroendocrine GLP-1R-responsive network, paralleling its central role in adult appetite regulation (Figure 1C).

Additionally, we investigated the identity of GLP-1R cells in the brain. GLP-1 and GDF15 regulate appetite by binding to GLP-1R (Glp1r) and GFRAL (Gfral) in the dorsal vagal complex. To better understand how these cells respond to ligands, we sought to understand transcriptomic differences between Glp1r and Gfral neurons. We used snRNA sequencing from the dorsal vagal complex (DVC) of 30 mice[14]. We subsetted all cells of the DVC to only cells expressing Glp1r, Gfral, or both and performed clustering by cell type to identify cells of interest. The source publication clustered cells by three layer identities, and we used these identifiers to understand the cell types in our subset of Glp1r+ and Gfral+ cells. Our subset did not include all of the cell types from the original dataset. Unsurprisingly, it was primarily composed of neurons and did not contain any connective or vascular tissue. Interestingly, *Sall3* neurons showed Glp1r expression but not Gfral expression. Previous studies have found that *Sall3* is required for the development of nerve projections from the hindbrain that innervate oral structures[18]. The presence of Glp1r in *Sall3* neurons is an important connection between sensory nerves related to taste and the medulla, which regulates appetite. Although this finding is preliminary, it indicates an interesting direction for future research regarding Glp1r and appetite-related sensory information. Additionally, we found Glp1r expression to be highest in the monoamine class of neurons, which synthesize norepinephrine and serotonin. Previous studies have demonstrated GLP-1 receptor agonist (GLP-1RA) administration can release serotonin, and there is a known relationship between serotonin and food intake[19]. As such, these medications may have additional pharmaceutical benefits related to mental health, in addition to metabolic disorders. Further investigation is needed to determine if GLP-1RA medications are safe and effective for the treatment of psychological illness[1].

GLP-1 and GDF15 share some similarities related to appetite reduction and body weight, but they are released by the body in very different contexts. While GLP-1 is secreted throughout the day, particularly in relation to meals[2],GDF15 levels rise in response to physiological stress, ranging from pregnancy to cancer. Under these conditions, GDF15 reduces appetite, and like GLP-1, binds to specific receptors in the dorsal vagal complex of the hindbrain. Considering the similarities and differences between GLP-1 and GDF15, we sought to identify transcriptomic similarities and differences between their receptor cells. Using a single nucleus RNA sequencing dataset of the mouse dorsal vagal complex, we compared cells with the receptor for GLP-1 (Glp1r) and cells with the receptor for GDF15 (Gfral). Surprisingly, differential expression analyses revealed that these cells are very similar, with very few differentially expressed genes.

Among the 9 differentially expressed genes across cell types expressing Glp1r or Gfral, we were most interested in *SLC6A2*. *SLC6A2* is also known as the Norepinephrine Transporter Gene, and it is related to psychiatric disorders including depression[20]. Prior research has identified that GLP1-RA medications may increase neurotransmitter release and synaptic plasticity[1]. Thus, it is plausible that some of these effects may be related to norepinephrine in GLP-1R+ neurons. Given that Glp1r expression was highest in the monoamine neurons that produce norepinephrine and serotonin, this finding may further validate the relationship between GLP-1 medications and depression, which is an area of ongoing research.

When subsetted to only neurons expressing Glp1r or Gfral, we identified another DEG of interest between Glp1r+ and Gfral+ neurons. *Lcp2* was significantly upregulated in Gfral+ neurons. *Lcp2* upregulation in

Gfral+ neurons was of particular interest because both *Lcp2* and GDF15 are involved in immunity. Interestingly, *Lcp2* is required for T-cell development[21], and previous work has found that GDF15 is related to regulatory T-cell formation[13, 22]. These findings suggest that Gfral+ cells may be implicated in T-cell maintenance and development.

Gene set enrichment analysis suggested a relatively subtle pathway level difference between Glp1r+ and Gfral+ DVC populations. Across the full cell and neuron restricted comparisons, only the *Hallmark KRAS signaling up* gene set remained significant after Benjamini-Hochberg correction. This KRAS-related signal was directionally supported by a reciprocal nominal enrichment of *Hallmark KRAS signaling down* in Gfral+ cells, and by a weaker but similarly oriented pattern in the neuron only analysis.

The Hallmark KRAS signaling up gene set comprises genes induced downstream of activated KRAS and is generally interpreted as a broader Ras/MAPK responsive transcriptional program as opposed to evidence of a KRAS mutation itself. KRAS is a small GTPase that functions as a molecular switch downstream of growth factor receptors and signals primarily through the RAF-MEK-ERK cascade to regulate gene expression programs involved in signalling, survival, and cellular state transitions[23]. Although an important oncogene, recent work defining the KRAS and ERK-dependent transcriptome demonstrates that activation of this pathway induces a coordinated transcriptional program across diverse cell types rather than a cancer-specific signature alone[23]. Klomp et al argues that the Hallmark KRAS signature is inaccurate and that their proposed KRAS-ERK transcriptome better reflects true KRAS signaling. Exploring the authors' proposed transcriptome would be an interesting next step.

In our data, this signal was the strongest in the full cell comparison and only directionally preserved in the neuron restricted analysis, suggesting that it may not reflect a robust neuron specific program unique to Glp1r+ cells. One possible explanation is that the enrichment is amplified by broader cell state comparison in the gull DVC dataset. Although current DVC single-cell atlases detect Glp1r+ primarily in neuronal populations, Glp1r+ signaling has been shown to engage surrounding astrocytes in the NTS, which contribute to GLP-1-mediated feeding responses and may alter local transcriptional programs[26]. In contrast, Gfral expression is more anatomically restricted within the area postrema and corresponds to a narrower neuronal population involved in GF15-mediated aversive feeding suppression[27]. A second possibility is that Glp1r+ spans several AP and NTS neuronal populations, whereas Gfral+ lies in a more specific subset. Thus pooling across all cells accentuates a modest Ras/MAPK-like transcriptional program that becomes diluted after neuron only restriction.

Relative to Glp1r+ cells, Gfral+ cells showed nominal enrichment for inflammatory response, epithelial-mesenchymal transition, angiogenesis, WNT/beta-catenin signaling, and xenobiotic metabolism. These terms were driven by leading-edge genes such as Postn, Col1a1, Col5a2, Sparc, Timp1, Vcam1, Fgf2, Lef1, Dll1, and Notch1 pointing to extracellular matrix remodelling, vascular interface, and inflammatory programs. This pattern is plausible in the context of Gfral+ cells, many of which are associated with the area postrema and related DVC regions that are specialized for sending circulating aversive signals. In contrast, Glp1r+ cells showed nominal enrichment of complement interferon related programs, suggesting additional immune responsive or stress responsive features, although these did not survive multiple testing correction and thus should be interpreted with caution.

Although our analysis of snRNA sequencing data of the DVC did not reveal many significant differences between Glp1r+ and Gfral+ cells, we did find interesting relationships between these cells and their possible functions. Our analysis of Glp1r+ and Gfral+ neurons in the dorsal vagal complex revealed a link between Glp1r and mental health-related neurotransmitters, including norepinephrine and serotonin. We first see that Glp1r expression is highest in neurons that synthesise these molecules, and we also found that the norepinephrine transporter gene is upregulated in Glp1r+ neurons relative to Gfral+ neurons. These point to the study of Glp1r agonists as a potential therapeutic for depression. Additionally, we discovered a link between *Sall3* neurons and Glp1r, which may establish a relationship between sensory information in the mouth and appetite-sensing neurons in the hindbrain, a relationship that is biologically

reasonable. Finally, we found a consistent relationship between Gfral+ cells and T-cell related genes, suggesting another area of future work to investigate the relationship between GDF15 and immunity, particularly in the context of pregnancy and cancer.
One major limitation of our investigation was that these datasets included relatively few Glp1r+ cells. For Dataset 1, pseudobulk expression profiles aggregate signals across all cells within a type, which improves detection of low-expression genes like Glp1r that are subject to severe dropout in single-cell data, but masks cell-to-cell heterogeneity. In addition, CytoTRACE2 and PAGA were performed on 60 cells per type, which was sufficient for broad potency ranking but limited any subgroup analysis. Finally, all developmental analyses were performed in mouse prenatal tissue and require validation in human systems.

Methods

GLP-1R Cell-Type Signature Analysis

We ranked cell types in a downsampled pilot atlas by the fraction of Glp1r+ cells and selected the top three eligible populations for within cell-type modeling. For each cell type, Glp1r+ cells were compared with matched Glp1r- cells sampled at a 1:2 ratio after stratification by day, experimental ID, UMI count, and gene count. Count matrices were log1p-transformed, filtered to genes detected in at least 1% of cells, and the Glp1r feature itself was excluded. Feature selection was performed using repeated L1-regularization logistic regression and benchmarked against XGboost. Stable L1 features were defined as genes with nonzero coefficients in at least 60% of repeats, whereas stable XGBoost features were refined as genes ranking among the top 100 permutation-important features in at least 60% of repeats. Model performance was assessed using repeated stratified cross validation and summarized by AUROC and AUPRC.

Clustering of DVC Glp1r+ and Gfral+ cells

We used R v 4.5.0 for all differential expression analysis of the DVC dataset acquired from Hes et al. 2025 (hereafter referred to as DVC dataset). We downloaded the publicly available mouse DVC dataset as an h5ad file from the Broad Institute Single Cell Portal. We used the zellkonverter and Seurat 4.5.0 package in R to convert the h5ad file to a Seurat object and save as an RDS file. We then performed quality control on this dataset by looking for mitochondrial gene expression. All mitochondrial gene expression fell between 0% and 3%, with the vast majority falling below 1%, indicating that low quality cells had already been filtered out of this dataset. We then subsetted this 6.5 GB Seurat object to only cells that express Glp1r, Gfral, or both above a threshold of 0. We saved this as a new RDS file, which we used for further analyses. We then used NormalizeData in Seurat to normalize the counts of our subset of cells. We then used FindVariableFeatures and ScaleData on our subset and used RunPCA to complete a principle component analysis. Using ElbowPlot, we decided to use 6 as our number of PCs and used the dimensions 1:6 for FindNeighbors. After trial and error, we chose a resolution of 0.15 for FindClusters. We then used RunUMAP in Seurat to complete the dimensionality reduction. We used DimPlot to construct a UMAP. To determine the cellular identities of the clusters, we used the clustering classes from the original DVC dataset and validated these using FindAllMarkers in Seurat. Clusters were assigned based on the majority of cells (greater than 50%) expressing genes related to the assigned cell type. All clusters had over 70% of cells positive for the cell type, with the exception of astrocytes, which were 36% of the cluster and Golli-related cells (which express glial markers) making up the other 14%.

Differential expression between Glp1r+ and Gfral+ cells

For our differential expression analysis between Glp1r+ and Gfral+ cells, we used FindMarkers in Seurat. We did not include cells that express both receptors in this analysis. We used EnhancedVolcano in R to visualize our differentially expressed genes. To investigate neurons in our subset, we created a new RDS

file by subsetting our Glp1r+ and Gfral+ cell population to only neurons based on the cell identities in the original metadata. For differential expression analysis, our significance requirements were adjusted p-value < 0.05 and absolute value of average $\log_2$FoldChange > 1.

To place the differential expression results into a border biological context, we performed gene set enrichment analysis (GSEA), cell composition analysis using neuronal cluster proportions, and gene co-expression analysis comparing Glp1r+ and Gfral+ populations. Here, GSEA allows us to assess whether transcriptional programs, as opposed to individual genes, were distinguishing Glp1r+ and Gfral+ signatures. Co-expression analysis helped uncover genes more likely to change expression with either Glp1r+ or Gfral+.

As this built off of the DEG analysis performed earlier, this was done at both the full-cell-population and neuron-restricted levels. More specifically, GSEA was performed on a preranked gene list derived from the same Seurat differential expression analysis, with genes ranked by average $\log_2$FoldChange. Enrichment was evaluated using the MSigDB Hallmark collection[24], a curated set of 50 gene sets that captures major biological processes while reducing reduction among overlapping pathways.

Pseudobulk expression profiling

Pre-aggregated expression files from the Qiu et al. mouse prenatal atlas were used to quantify mean expression and percent cells expressing for each gene per cell type per developmental timepoint. We focused on four genes: Glp1r, Gcgr, Gipr, and Gcg. For time-course analyses, a cell type was counted as expressing a gene if at least 1% of cells in that cell type at that timepoint were positive. Extraembryonic visceral endoderm was excluded in some analyses because it is not part of the embryo proper.

CytoTRACE2 developmental potency scoring

Single-cell raw counts for 12 curated GLP-1R-relevant cell types were obtained from CellxGene Census[25]. We sampled 60 cells per cell type (720 total). CytoTRACE2[10] was run on raw counts to generate potency scores ranging from 0 to 1, where lower values indicate greater differentiation and higher values indicate greater developmental potency.

PAGA trajectory inference

For trajectory analysis, we subset to 6 neuroendocrine cell types: hypothalamus cell, pituitary gland cell, enteric neuron, parasympathetic neuron, olfactory epithelial cell, and cochlear auditory hair cell (360 cells total). Cells were analyzed using a standard Scanpy workflow with neighborhood graph construction, UMAP embedding, and partition-based graph abstraction (PAGA)[11]. The broader 12-cell-type panel was not used for PAGA because it included biologically unrelated tissues.

Code Availability

https://github.com/rnayebi21/205_proj

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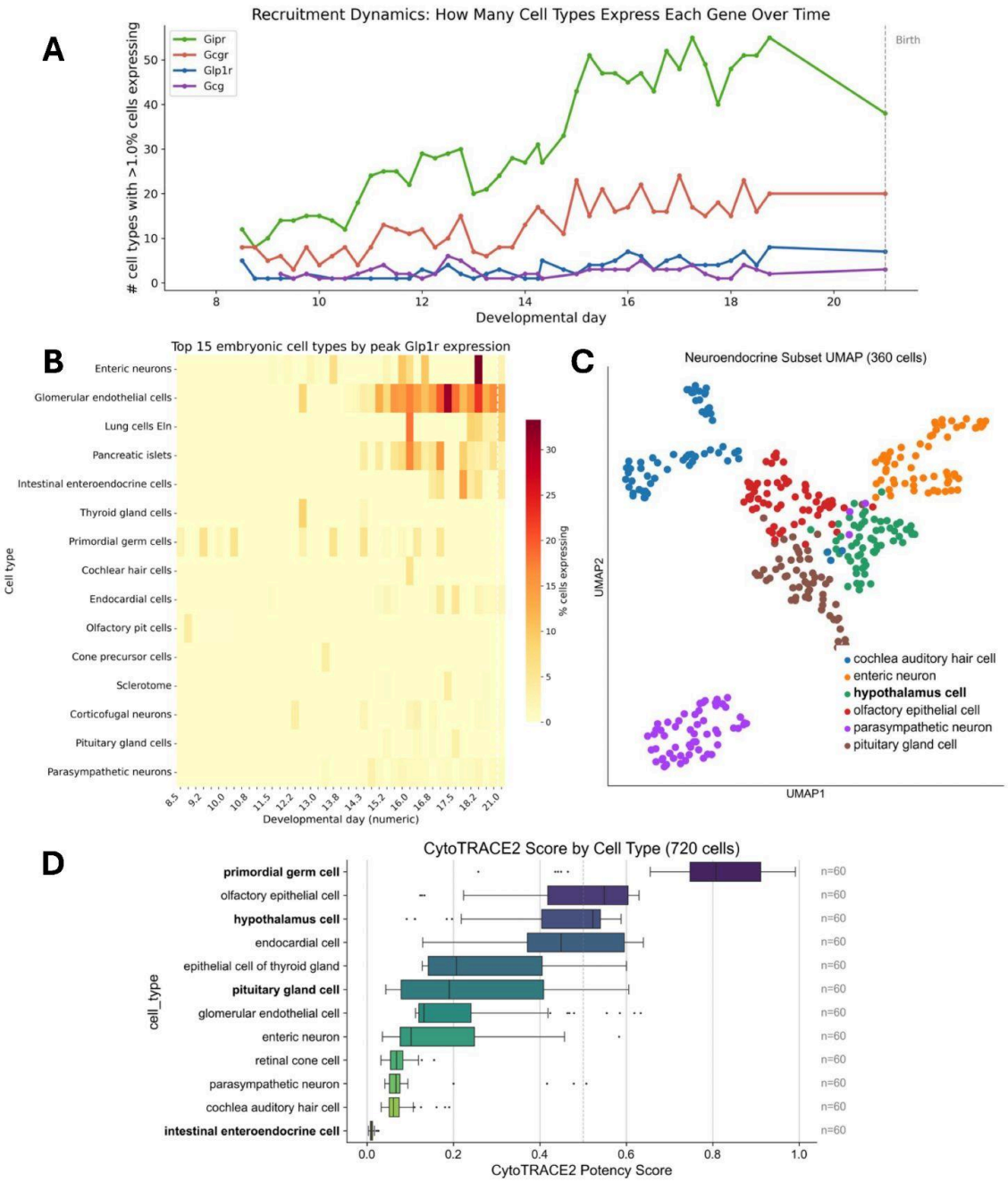

Figure 1: Developmental profiling of the incretin axis across mouse prenatal development. A) Recruitment dynamics showing the number of cell types expressing each gene above 1% threshold across developmental time. Gipr (green) is expressed in up to 55 cell types by mid-gestation, while Glp1r (blue) remains between 2 and 8. B) Heatmap of Glp1r expression (% cells expressing) across cell types and developmental time, with extraembryonic visceral endoderm excluded. Dark blocks indicate expression concentrated in specific late-gestation windows. C) PAGA trajectory analysis of the neuroendocrine subset (6 cell types, 360 cells). Hypothalamus cell is the central hub with the strongest connectivity to olfactory epithelial cell (0.40) and pituitary gland cell (0.23). D) CytoTRACE2 developmental potency scores for 12 GLP-1R-relevant cell types (60 cells per type). Primordial germ cells score highest (positive control); intestinal enteroendocrine cells (GLP-1 producers) are the most differentiated; hypothalamus and pituitary cells (GLP-1 responders) retain intermediate potency.

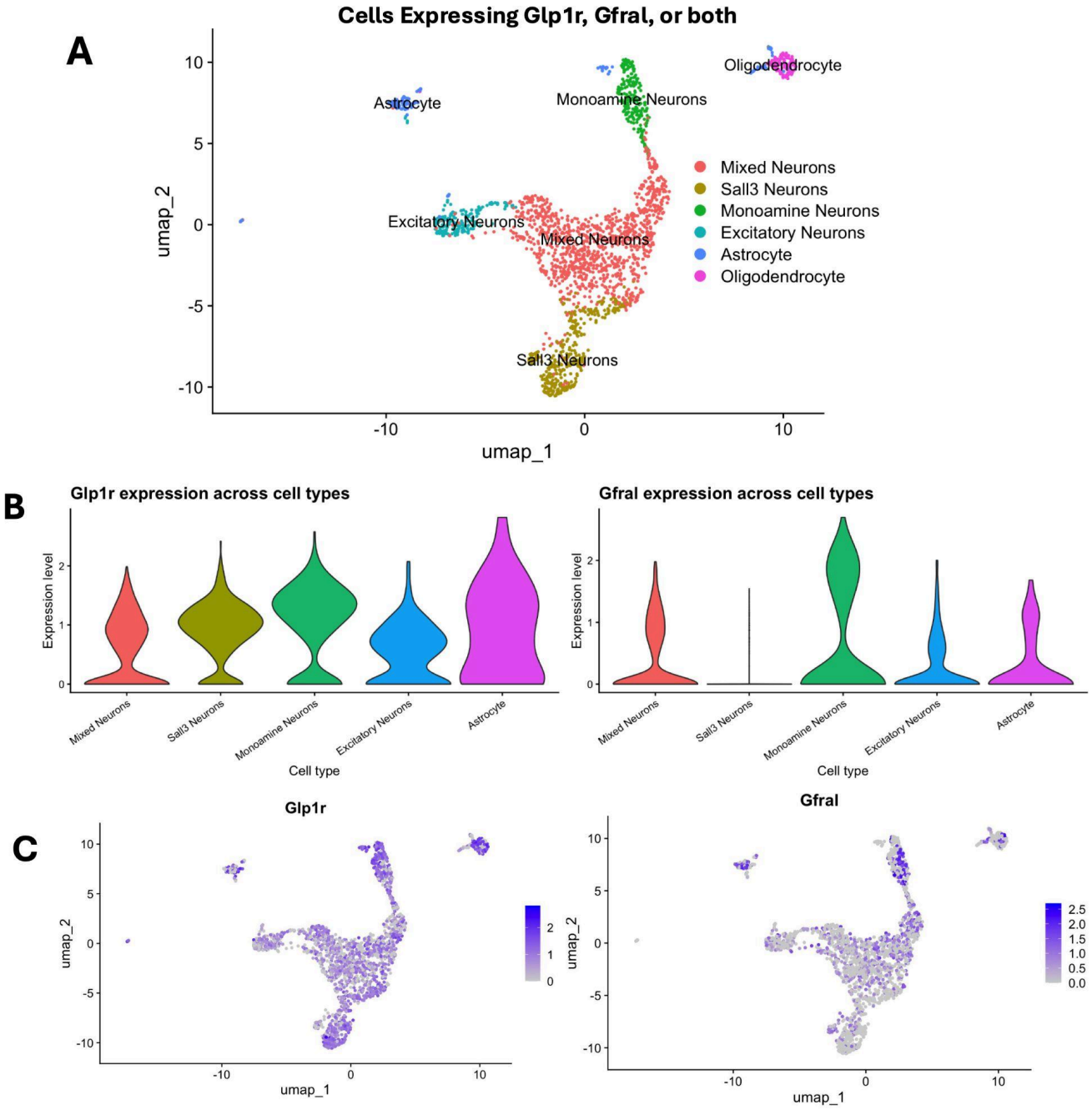

Figure 2: UMAP Clustering of Glp1r+ and Gfral+ cells in the dorsal vagal complex from Hes et al. A) UMAP clusters by cell types expressing Glp1r and/or Gfral. We identified 6 clusters in our DVC subset using Seurat and labeled them by cell type according to the original DVC publication (Cite). **B)** Glp1r and Gfral expression across cell types in our subset of DVC cells. **C)** Glp1r and Gfral expression across cell type clusters on the UMAP.

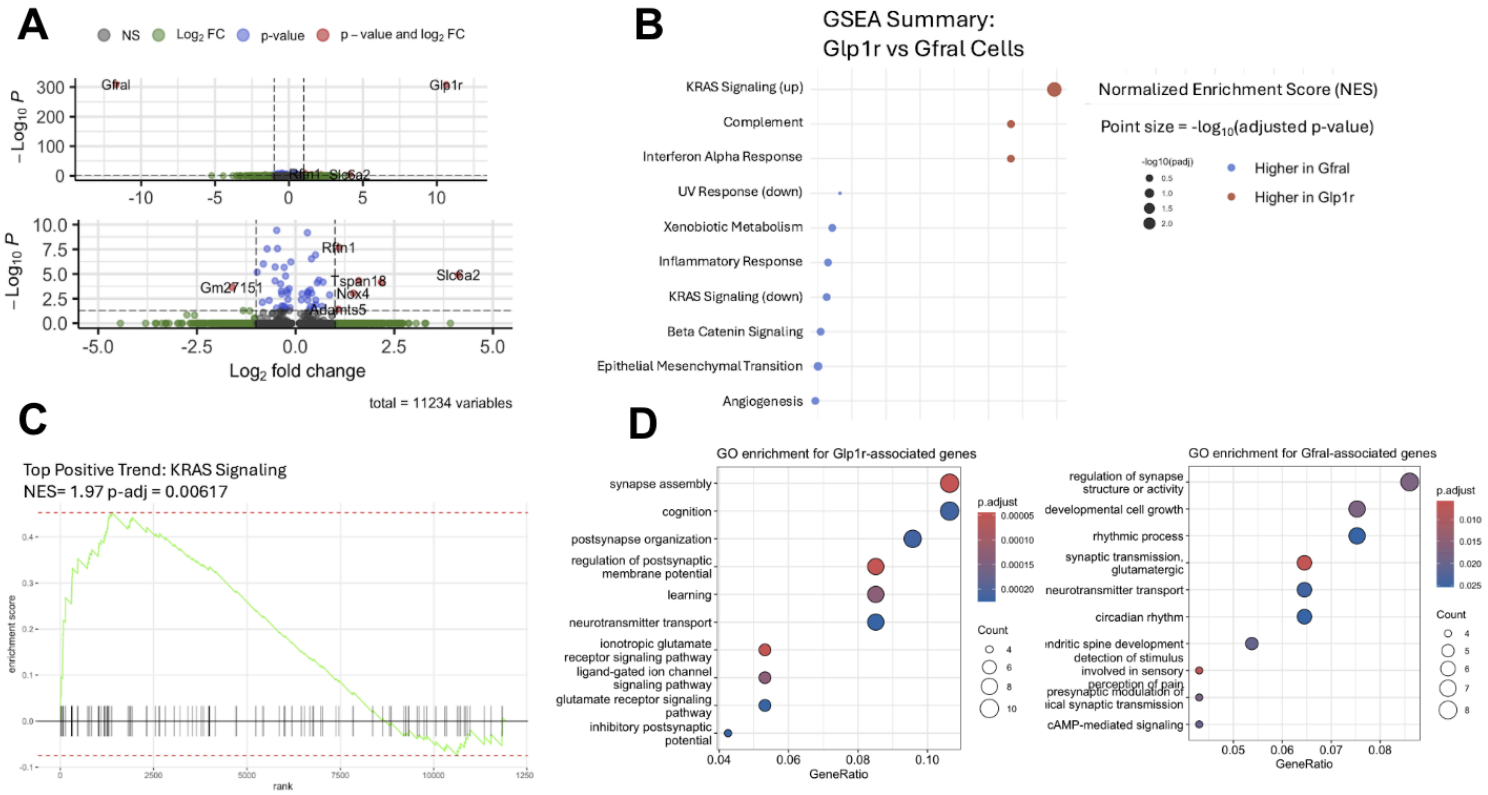

Figure 3: Differential expression analysis and Gene Set Enrichment Analysis between Glp1r+ and Gfral+ cells in the DVC. A) Volcano plot of differentially expressed genes between Glp1r+ versus Gfral+ cells based on $\log_2$FoldChange using Seurat and Enhanced Volcano. **B)** Bubble plot summarizing the top 10 Hallmark gene sets ranked by BH adjusted p-values identified by preranked GSEA in all Glp1r+ versus Gfral+ cells. The *x*-axis represents normalized enrichment score (NES). Bubble size represents -log10 of adjusted p-value, and color denotes the direction of Enrichment. *Hallmark KRAS signaling up* was the only significant gene set. **C)** GSEA enrichment curve for the Hallmark KRAS signalling up gene set, the top positively enriched and only significantly enriched Hallmark gene set in all cell Glp1r+ versus Gfral+ comparison. **D)** Functional plots of the top 100 co-expressed DEGs between Glp1r+ and Gfral based on Spearman correlation scores.

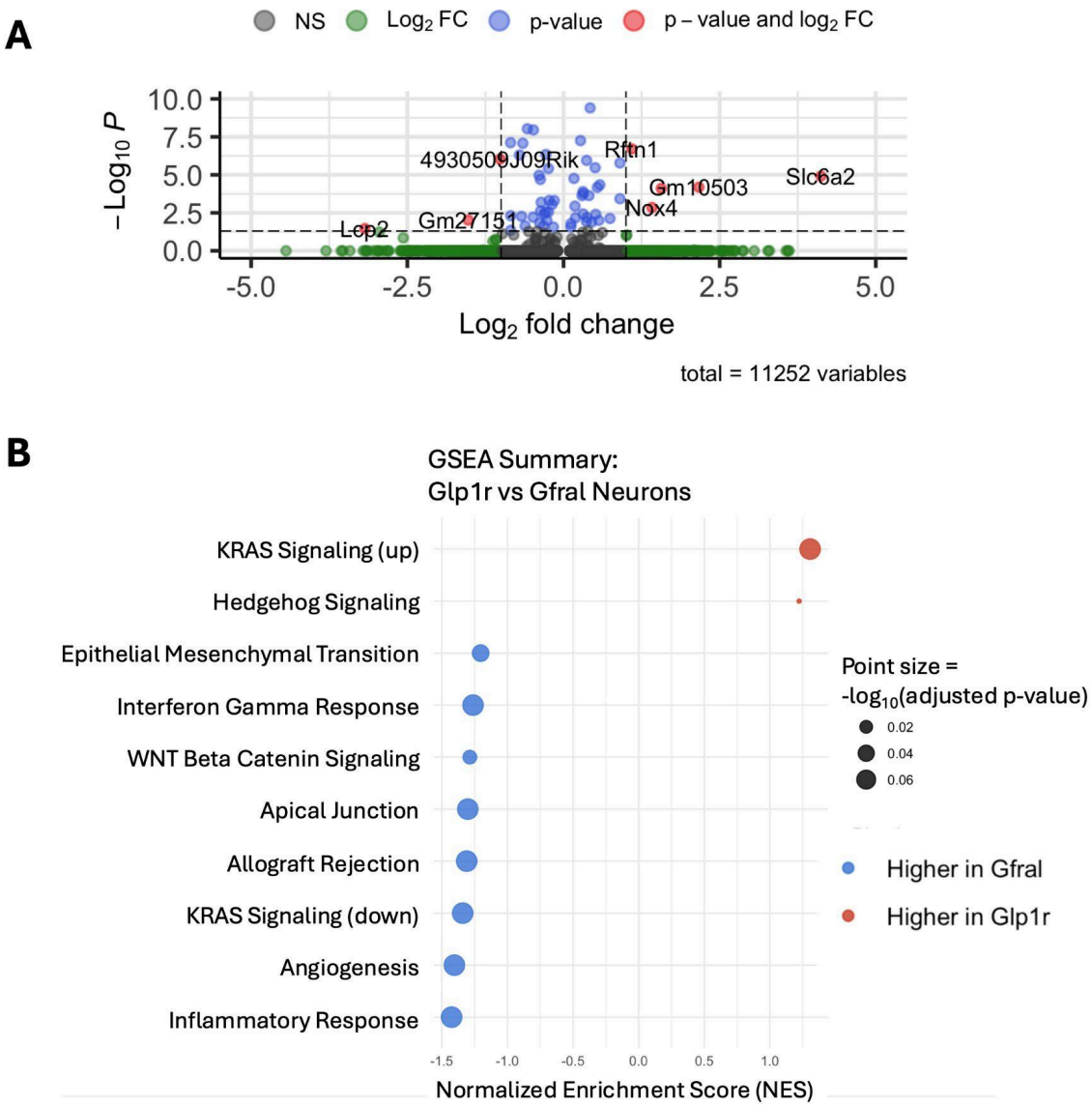

Figure 4: Differential expression analysis and Gene Set Enrichment Analysis between Glp1r+ and Gfral+ neurons in the DVC. A) Volcano plot of differentially expressed genes between Glp1r+ versus Gfral+ neurons based on $\log_2$FoldChange using Seurat and Enhanced Volcano. **B)** Bubble plot summarizing the top 10 Hallmark gene sets ranked by BH adjusted p-values identified by preranked GSEA in all Glp1r+ versus Gfral+ neurons. The *x*-axis represents normalized enrichment score (NES). Bubble size represents -log10 of adjusted p-value, and color denotes the direction of Enrichment. No significant gene sets were found.